

Dialysis: DTMF Signal Analysis Studio

Algorithms, Signal Processing, and Real-Time Decoding

September 24, 2026

Antigravity

What is DTMF?

Dual-Tone Multi-Frequency (DTMF) is the global standard for telecommunication signaling over analog telephone lines in the voice-frequency band.

How it works:

- Each key press generates **two** simultaneous pure sine waves.
- One from a *low-frequency* group (rows).
- One from a *high-frequency* group (columns).

	1209 Hz	1336 Hz	1477 Hz	1633 Hz
697 Hz	1	2	3	A
770 Hz	4	5	6	B
852 Hz	7	8	9	C
941 Hz	*	0	#	D

Tone Synthesis & Export

Step 1: Signal Generation

A DTMF signal $x[n]$ is simply the superposition of two sinusoids:

$$x[n] = \sin\left(2\pi f_{\text{low}} \frac{n}{f_s}\right) + \sin\left(2\pi f_{\text{high}} \frac{n}{f_s}\right)$$

where f_s is the sampling rate (typically 8000 Hz).

Step 2: Sequence Assembly

- To transmit a phone number (e.g., "512"), individual tones are generated and concatenated.
- Silent gap buffers (zeros) are inserted between tones to allow the receiver to segment discrete digits.

Signal Segmentation

Goal: Isolate active tone regions from continuous recordings (which may contain silence or room noise).

Algorithm:

1. **Framing:** Divide the full audio array into 10 ms chunks.
2. **RMS Energy:** Calculate the Root Mean Square (RMS) energy for each chunk.
3. **Thresholding:** Set a threshold $T = 0.05 \times \max(\text{RMS})$ to separate active tones from the noise floor.
4. **Merging:** Track consecutive loud chunks as a single contiguous region $\rightarrow [start, end]$.

Note: False positives picked up by the low threshold are discarded later by strict spectral validation.

The Goertzel Algorithm

Why not FFT? Computing an entire FFT is wasteful if we only care about 8 specific frequencies.

The Math: The Goertzel algorithm computes a single-bin DFT via a 2nd-order IIR filter. For a target frequency f_k :

$$\omega_k = 2\pi \frac{f_k}{f_s} \Rightarrow \text{Coeff} = 2 \cos(\omega_k)$$

Recurrence Relation:

$$s[n] = x[n] + \text{Coeff} \cdot s[n-1] - s[n-2]$$

Power Output:

$$|X[k]|^2 = s[N-1]^2 + s[N-2]^2 - \text{Coeff} \cdot s[N-1] \cdot s[N-2]$$

The target frequencies map to conjugate pole pairs $e^{\pm j\omega_k}$ on the unit circle.

Custom Fast Fourier Transform (FFT)

Implementation: Radix-2 Cooley-Tukey Algorithm in pure Python.

- **Zero-Padding:** Input is padded to the next power of 2.
- **Bit Reversal:** Input array indices are sorted in bit-reversed order.
- **Butterfly Operations:** Computed iteratively across $\log_2(N)$ stages.
- **Twiddle Factors:** $W_N^k = e^{-j\frac{2\pi}{N}k}$ applied at each level.

While computationally $O(N \log N)$ like standard implementations, running it in pure Python over long live audio buffers poses a performance challenge (solved by isolating analysis to bounded trailing windows).

Enhanced FFT Decoding

Raw rectangular-window FFT peak-picking is prone to spectral leakage and bin resolution issues. Dialysis uses an enhanced FFT path:

1. **Hann Windowing:** Multiplies the tone region by a Hann window to kill spectral leakage.
2. **Zero-Padding (2x):** Artificially increases frequency domain bin resolution.
3. **Parabolic Peak Interpolation:** Finds the true sub-bin peak by fitting a parabola to the highest magnitude bin and its two neighbors.
4. **Tolerance Check:** Snaps the estimated peak to the nearest nominal DTMF frequency (must be within $\pm 2.5\%$).

Live Decoding Engine (DtmfTracker)

Static file thresholds fail on live microphones due to dynamic room noise. We use a **Temporal State Machine**.

- **Frame Geometry:** Audio is analyzed in 20 ms frames with a 5 ms hop size.
- **Adaptive Noise Floor:** Min-with-leak tracker. Drops instantly, rises max 10 dB/s.
- **Vectorized Goertzel:** Computes 8 bin powers per frame via a single matrix multiplication.
- **Latch & Release:**
 - Latches a digit after 6 consecutive agreeing frames (30 ms).
 - Releases after 4 consecutive disagreeing frames (20 ms).

ITU-T Q.24 Validation

Once a tone is latched, it must pass rigorous telecommunication standards to be accepted as genuine DTMF (rejecting speech, music, and noise).

Validation Checks:

1. **Dial Tone:** The 350+440 Hz PSTN dial tone pair must not dominate.
2. **Twist (Amplitude Ratio):**
 - Forward Twist: Row tone max 15 dB louder than column tone.
 - Reverse Twist: Column tone max 12 dB louder than row tone.
3. **Frequency Offset:** The identified frequency must be the local maximum among $\pm 2.5\%$ neighborhood probes.
4. **Harmonic Drop:** The 2nd harmonic of either tone must be ≥ 6 dB below its fundamental (rejects human voice).

Noise & Aliasing (Phase 10)

Dialysis provides active controls to deliberately break the decoders:

Signal-to-Noise Ratio (SNR)

- Injects Additive White Gaussian Noise (AWGN) to a specific SNR limit.
- Demonstrates decoder resilience (the adaptive gate survives down to ~ 0 dB SNR).

Nyquist-Shannon Aliasing

- Resampling the signal below $f_s = 3267$ Hz (where $f_{Nyquist} = 1633.5$ Hz) introduces destructive frequency folding.
- High-frequency column tones alias into the lower spectrum, permanently destroying the DTMF signature.

Conclusion

Dialysis successfully bridges textbook DSP concepts with robust real-time software engineering.

Thank You!